AQUEN FLUID TECHNOLOGIES LLP

Complete Water & Wastewater Treatment Solutions

Design | Manufacturing | Supply | Installation | Commissioning | AMC

ANION BED

DEMINERALISATION (DE-IONISATION) SYSTEM

REMOVAL OF ANIONS FOR HIGH PURITY WATER

FRP / MSRL / MSEP VESSELS | 100 LPH – 10,000 LPH

- High Purity Treated Water
- Low Operating Cost
- FRP, MSEP & MSRL Options
- Automatic / Manual Operation
- Suitable for Industrial Applications
- Easy Regeneration & Maintenance



FRP ANION BED
(100 LPH – 5000 LPH)
With Multiport Valve (MPV)



MSEP ANION BED
(1000 LPH – 10000 LPH)
MS with Epoxy Lining



MSRL ANION BED
(500 LPH – 10000 LPH)
MS with FRP Lining

PROCESS FLOW DIAGRAM – ANION BED SYSTEM



APPLICATIONS / INDUSTRIES



WORKING PRINCIPLE

Anion Bed contains Strong Base Anion Resin (SBA) which removes anions from water such as Chloride (Cl^-), Sulphate (SO_4^{2-}), Nitrate (NO_3^-), Silica (SiO_2), Bicarbonate (HCO_3^-) etc. The anions are exchanged with hydroxide ions (OH^-) and effluent after anion bed is demineralized water. The resin is regenerated using caustic soda (NaOH). It is generally used in combination with a cation bed for complete demineralisation (DM) system.



KEY FEATURES

- Removes anions and produces high purity water
- Available in FRP, MSEP and MSRL construction
- Flow range: 100 LPH to 10,000 LPH
- Low pressure drop and high exchange capacity
- High quality ion exchange resin
- Manual / Automatic operation options
- Compact and rugged design
- Easy regeneration with NaOH
- Suitable for continuous and intermittent operation
- Custom design as per outlet water quality requirement

INLET & OUTLET PARAMETERS (TYPICAL)

Parameter	Unit	Typical Inlet (from Cation/Pre-treated)	Typical Outlet (After Anion Bed)
pH	-	6.5 – 8.5	8.0 – 9.5
Total Dissolved Solids	mg/L	50 – 500	< 10 (with Cation + Anion) < 50 (only Anion Bed)
Chloride (Cl^-)	mg/L	< 50	< 1
Sulphate (SO_4^{2-})	mg/L	< 50	< 1
Nitrate (NO_3^-)	mg/L	< 10	< 1
Silica (SiO_2)	mg/L	< 20	< 1
Conductivity	µS/cm	100 – 1000	< 5 – 20

OPERATING PARAMETERS

Parameter	Value
Service Flow Rate	As per table
Operating Pressure	2.5 – 4.0 kg/cm²
Maximum Pressure	6.0 kg/cm²
Operating Temperature	5 – 45 °C
Bed Depth	800 – 1200 mm
Free Board	30 – 50%
Regeneration	4 – 8% NaOH
Regeneration Time	60 – 90 minutes
Slow Rinse	30 minutes
Fast Rinse	30 minutes
Operating Bed Volume (OBR)	10 – 40 m³/m³ (depending on water quality)

TECHNICAL SPECIFICATIONS – ANION BED (100 LPH TO 10,000 LPH)

Model	Flow Rate (LPH)	Vessel Type	Vessel Dia (mm)	Shell Height (mm)	Media Quantity (Ltr)	Service Flow Rate (m³/hr)	Backwash Flow Rate (m³/hr)	Inlet / Outlet Size (NB)	Operating Bed Volume (m³/m³)
AQU-AB-0.1	100	FRP	200	1200	25	0.1	0.3	15	10 – 25
AQU-AB-0.25	250	FRP	300	1400	50	0.25	0.6	20	10 – 30
AQU-AB-0.5	500	FRP	400	1650	75	0.5	1.2	25	10 – 30
AQU-AB-1.0	1,000	FRP / MSRL	500	1800	125	1.0	2.0	32	15 – 30
AQU-AB-2.0	2,000	FRP / MSEP	700	2000	250	2.0	4.0	40	15 – 35
AQU-AB-3.0	3,000	MSEP / MSRL	800	2200	400	3.0	6.0	50	15 – 35
AQU-AB-5.0	5,000	MSEP / MSRL	1000	2400	600	5.0	10.0	65	20 – 40
AQU-AB-7.5	7,500	MSEP / MSRL	1200	2600	900	7.5	15.0	80	20 – 40
AQU-AB-10	10,000	MSEP / MSRL	1400	2800	1200	10.0	20.0	100	20 – 40

Note: Above specifications are indicative and can be customized as per water quality and site conditions.

TYPICAL BILL OF MATERIAL (BOM)

Sr. No.	Item	Quantity
1	Anion Bed Vessel (FRP / MSEP / MSRL)	1 No.
2	Strong Base Anion Resin (Type I/II)	As per capacity
3	Multiport Valve (FRP) / Butterfly Valves (MSEP/MSRL)	1 Set
4	Top & Bottom Distribution System	1 Set
5	Piping (UPVC / CPVC / MS / SS - as required)	1 Set
6	Pressure Gauge	2 Nos.
7	Sample Valve	1 No.
8	Air Release Valve (ARV)	1 No.
9	Drain Valve	1 No.
10	Flow Meter (Optional)	1 No.
11	Control Panel (Auto / Manual) – Optional	1 No.
12	Chemical Dosing System for NaOH (Optional)	1 Set.
13	Regeneration Tank (Optional)	1 No.
14	Interconnecting Fittings, Supports & Accessories	1 Set

MATERIAL OF CONSTRUCTION (MOC)

Item	MOC Options
Vessel	FRP / MS (Epoxy Lined – MSEP) / MS (FRP Lined – MSRL)
Piping	UPVC / CPVC / PP / MS / SS-304 / SS-316
Valves	UPVC / CPVC / PP / CI / SS-304 / SS-316
Distribution System	UPVC / CPVC
Multiport Valve (FRP)	UPVC / ABS
Instrumentation	SS-304 / SS-316
Control Panel	MS with Powder Coating / SS-304 (Optional)
Resin	Strong Base Anion Resin (Type I or II)
Gaskets / O-rings	EPDM / Viton
Supports / Skid	MS with Powder Coating / SS-304

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WTP | STP | ETP | ZLD | FABRICATION | SPARES | AMC | RETROFIT | AUTOMATION

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SUSTAINABLE WATER SOLUTIONS FOR A BETTER TOMORROW

1. FRP ANION BED INSTALLATION – 2,000 LPH

SITE IN PROGRESS (FRP VESSEL WITH MPV)



2. FRP ANION BED SYSTEM – 500 LPH

INSTALLATION IN PROGRESS (SKID MOUNTED)



3. MSEP ANION BED – 5,000 LPH

SITE IN PROGRESS (MS VESSEL WITH EPOXY LINING)



4. MSRL ANION BED – 3,000 LPH

SITE IN PROGRESS (MS VESSEL WITH FRP LINING)



5. ANION BED – 1,000 LPH

INSTALLATION IN PROGRESS (FRP VESSEL WITH FRONT PIPING)



6. MSEP ANION BED – 10,000 LPH

SITE IN PROGRESS (MS VESSEL WITH EPOXY LINING)

